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for Physics of Nanostructures

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"Nonlinear Control Systems"

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1 SOLUTIONS

1.1 Example 2

For the following systems, show that the system has no limit cycles.

$$\dot{x}_1 = -x_1 + x_1^3 + x_1x_2^2,$$

$$\dot{x}_2 = -x_2 + x_2^3 + x_1^2x_2,$$

1.1.1 Solution

Define $V = x_1^2 + x_2^2$. Then:

$$\begin{aligned}\dot{V} &= 2x_1\dot{x}_1 + 2x_2\dot{x}_2 \\ &= 2x_1 [x_1(-1 + x_1^2 + x_2^2)] + 2x_2 [x_2(-1 + x_1^2 + x_2^2)] \\ &= 2(x_1^2 + x_2^2)(-1 + x_1^2 + x_2^2) \\ &= 2V(V - 1).\end{aligned}$$

The radial dynamics are governed by:

$$\dot{V} = 2V(V - 1).$$

Equilibrium points occur when $\dot{V} = 0$:

- $V = 0$ (the origin)
- $V = 1$ (the unit circle)

On the unit circle $V = 1$, we have:

$$\dot{x}_1 = x_1(-1 + 1) = 0, \quad \dot{x}_2 = x_2(-1 + 1) = 0.$$

Thus, every point on the unit circle is an equilibrium.

For $0 < V < 1$: $\dot{V} < 0$, trajectories approach the origin.

For $V > 1$: $\dot{V} > 0$, trajectories move away from the unit circle.

Since the unit circle consists entirely of equilibria and the flow is monotonic elsewhere, no nontrivial limit cycles exist.

1.2 Example 3

Consider the system:

$$\begin{aligned}\dot{x}_1 &= -1 - x_1x_2, \\ \dot{x}_2 &= x_1,\end{aligned}$$

1.2.1 Solution

Assume a periodic solution exists with period $T > 0$.

From $\dot{x}_2 = x_1$, integrate over one period:

$$\int_0^T \dot{x}_2 dt = x_2(T) - x_2(0) = 0 = \int_0^T x_1(t) dt. \quad (1)$$

From $\dot{x}_1 = -1 - x_1x_2$, integrate over one period:

$$\int_0^T \dot{x}_1 dt = x_1(T) - x_1(0) = 0 = \int_0^T [-1 - x_1(t)x_2(t)] dt. \quad (2)$$

Right side:

$$-\int_0^T 1 dt - \int_0^T x_1(t)x_2(t) dt = 0 \Rightarrow -T - \int_0^T x_1(t)x_2(t) dt = 0. \quad (3)$$

since that:

$$\int_0^T x_1(t)x_2(t) dt = \int_0^T x_2(t)\dot{x}_2(t) dt = \int_0^T \frac{1}{2} \frac{d}{dt}(x_2^2) dt = \frac{1}{2}[x_2^2(T) - x_2^2(0)] = 0.$$

Substitute into (3):

$$-T - 0 = 0 \Rightarrow T = 0.$$

No periodic solutions exist.

1.3 Example 4

Consider the system:

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= -[2b - g(x_1)]ax_2 - a^2x_1,\end{aligned}$$

where $a > 0, b > 0$

1.3.1 Solution

The divergence of the vector field is:

$$\nabla \cdot \mathbf{f} = \frac{\partial}{\partial x_1}(x_2) + \frac{\partial}{\partial x_2}(-a[2b - g(x_1)]x_2 - a^2x_1) = -a[2b - g(x_1)].$$

In particular:

- If $g(x_1)$ is bounded: exists b such that $2b - g(x_1) > 0$ for all x_1 , then $\nabla \cdot \mathbf{f} < 0$ everywhere, and hence no limit cycles exist in the plane.
- If $g(x_1)$ causes $2b - g(x_1)$ to change sign, then the criterion is inconclusive, and limit cycles may potentially exist.

1.4 Example 5

Consider the system:

$$\begin{aligned}\dot{x}_1 &= x_1 + x_2 - x_1(x_1^2 + x_2^2) \\ \dot{x}_2 &= -2x_1 + x_2 - x_2(x_1^2 + x_2^2)\end{aligned}$$

Find the periodic orbits.

1.4.1 Solution

Introduce polar coordinates $x_1 = r \cos \theta$, $x_2 = r \sin \theta$ with $r \geq 0$. A direct computation (or using $r\dot{r} = x_1\dot{x}_1 + x_2\dot{x}_2$ and $r^2\dot{\theta} = x_1\dot{x}_2 - x_2\dot{x}_1$) gives

$$\dot{r} = r\left(1 - \frac{1}{2}\sin(2\theta) - r^2\right), \quad \dot{\theta} = -(1 + \cos^2 \theta) < 0.$$

Since $\dot{\theta} < 0$ for all θ , θ is strictly monotone and every non-equilibrium trajectory makes continuous clockwise rotations. For small r we have $\dot{r} \approx r > 0$ (radii grow), while for large r we have $\dot{r} \approx -r^3 < 0$ (radii decay). Thus trajectories starting at small radii move outward and those at large radii move inward.

Maybe there is one periodic orbit, lying in the annulus $1/\sqrt{2} \leq r \leq \sqrt{3/2}$

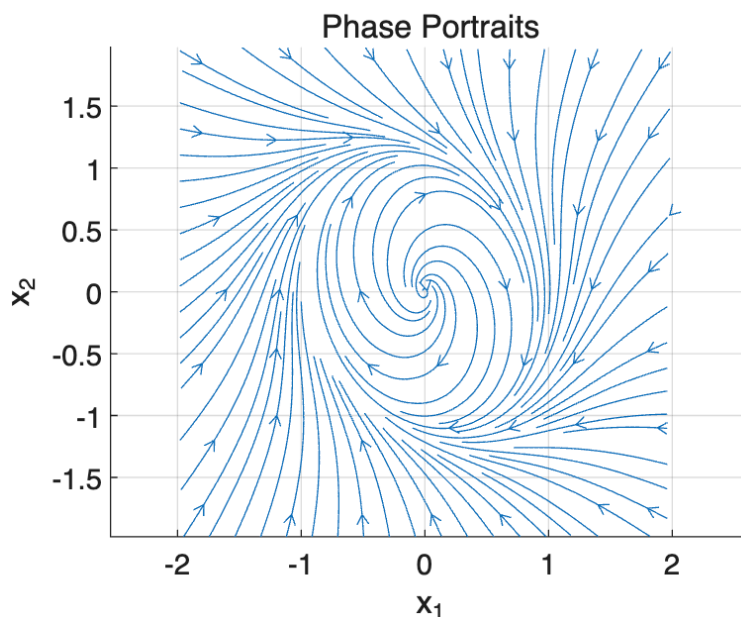


Figure 1 — Phase portrait

1.5 Example 7

Find the coefficients $g(a)$ and $b(a)$ for the following nonlinearities:

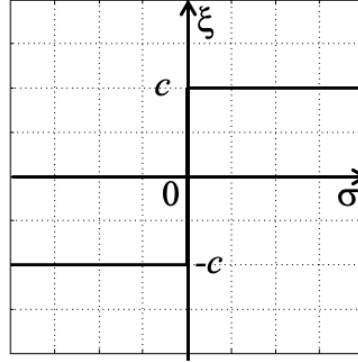


Figure 2 — Relay

1.5.1 Solution

From Example 6, we have:

$$g(a) = \frac{2c}{\pi a} \left(\sqrt{1 - \left(\frac{mb}{a}\right)^2} + \sqrt{1 - \left(\frac{b}{a}\right)^2} \right) \quad (1)$$

$$b(a) = -\frac{2cb(1-m)}{\pi a^2} \quad (2)$$

In this case, we have $m = 0$, $b = 0$, and $M = c$. Thus,

$$g(a) = \frac{2c}{\pi a} \int_{\psi_1}^{\psi_2} \sin \psi d\psi = -\frac{2c}{\pi a} \cos \psi \Big|_{\psi_1}^{\psi_2} = \frac{4M}{\pi a} = \frac{4c}{\pi a}, \quad (a > 0). \quad (3)$$

$$b(a) = \frac{2c}{\pi a} \int_{\psi_1}^{\psi_2} \cos \psi d\psi = \frac{2c}{\pi a} \sin \psi \Big|_{\psi_1}^{\psi_2} = 0 - 0 = 0 \quad (4)$$

1.6 Example 8

Find the coefficients $g(a)$ and $b(a)$ for the following nonlinearities:

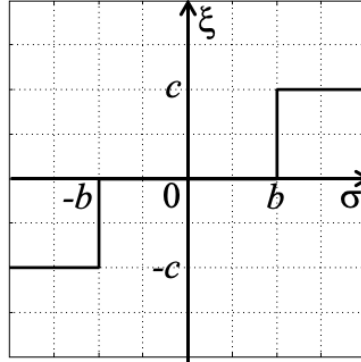


Figure 3 — Relay with dead zone

1.6.1 Solution

From Example 6, we have:

$$g(a) = \frac{2c}{\pi a} \left(\sqrt{1 - \left(\frac{mb}{a} \right)^2} + \sqrt{1 - \left(\frac{b}{a} \right)^2} \right) \quad (5)$$

$$b(a) = -\frac{2cb(1-m)}{\pi a^2} \quad (6)$$

In this case, we have $m = 1$, and $M = c$. Thus,

$$g(a) = \frac{2c}{\pi a} \left(\sqrt{1 - \left(\frac{b}{a} \right)^2} + \sqrt{1 - \left(\frac{b}{a} \right)^2} \right) = \frac{4c}{\pi a} \sqrt{1 - \left(\frac{b}{a} \right)^2} \quad (7)$$

$$b(a) = -\frac{2cb(1-m)}{\pi a^2} = 0 \quad (8)$$

1.7 Example 9

Find the coefficients $g(a)$ and $b(a)$ for the following nonlinearities:

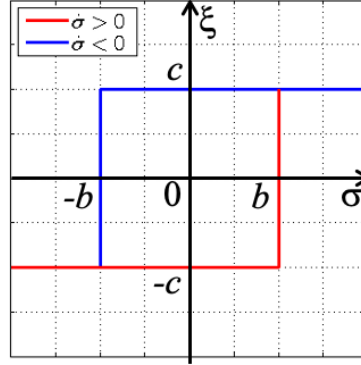


Figure 4 — Hysteresis

1.7.1 Solution

From Example 6, we have:

$$g(a) = \frac{2c}{\pi a} \left(\sqrt{1 - \left(\frac{mb}{a}\right)^2} + \sqrt{1 - \left(\frac{b}{a}\right)^2} \right) \quad (9)$$

$$b(a) = -\frac{2cb(1-m)}{\pi a^2} \quad (10)$$

In this case, we have $m = -1$, and $M = c$. Thus,

$$g(a) = \frac{2c}{\pi a} \left(\sqrt{1 - \left(\frac{b}{a}\right)^2} + \sqrt{1 - \left(\frac{b}{a}\right)^2} \right) = \frac{4c}{\pi a} \sqrt{1 - \left(\frac{b}{a}\right)^2} \quad (11)$$

$$b(a) = -\frac{4cb}{\pi a^2} \quad (12)$$